

操作系统原理

第一章：介绍

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目录

- 1 课程简介
 - 主要内容及参考资料
 - 为什么要学习操作系统原理？
- 2 What's an Operating System?
 - Components of a Computer System
 - What's an Operating System?
 - Components of an operating system
 - History of Operating System
 - Features migration

Outline

1 课程简介

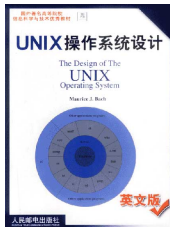
- 主要内容及参考资料
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2 What's an Operating System?

- Components of a Computer System
- What's an Operating System?
- Components of an operating system
- History of Operating System
- Features migration

课程简介

- 主要内容
 - 操作系统的基本概念、原理和设计。
- 教材及参考资料
 - **Abraham Silberschatz, Operating System Concepts, 6th Edition, John Wiley & Sons, Inc.**
 - 9th edition is also available.
 - Andrew S. Tanenbaum, Modern operating System, 2nd Edition, Prentice Hall.
 - Maurice J. Bach, The design of Unix operating system, Pearson education.



为什么要学习操作系统原理？

- 操作系统是最重要的系统软件之一，
 - 没有操作系统的计算机几乎不能使用；
 - 学习操作系统将帮助软件开发人员开发出优秀的应用软件系统。
- 操作系统是最复杂的系统软件之一，
 - 学习计算机系统是如何工作的；
 - 学习如何通过**抽象**来掌握复杂的软件系统；
 - 学习如何系统地设计复杂的软件系统。

Outline

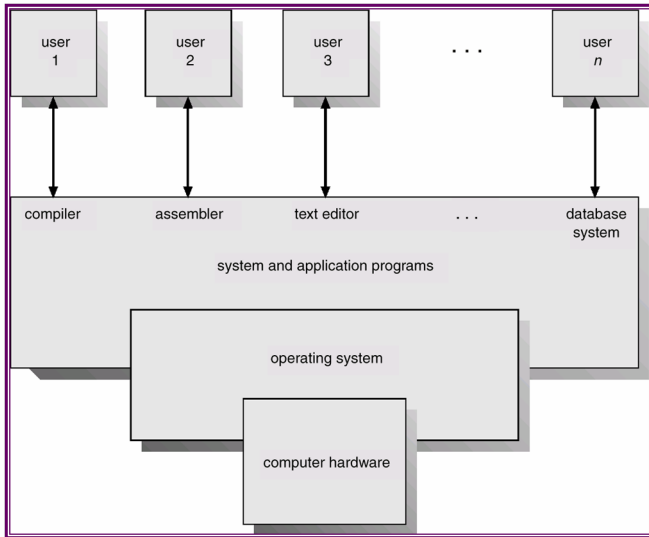
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Components of a Computer System



What's an Operating System?

- Different views
 - The operating system controls and coordinates the use of the hardware among the various application programs for the various users.
 - **Control program**
 - The operating system manages the resources of a computer so that various applications and users can operate the computer system efficiently and fairly.
 - **Resource manager (Resource allocator)**
 - The operating system abstracts the computer hardware and presents the user a friendly interface.
 - **Extended machine (Virtual machine)**
- So, what's an operating system?
 - **No** common accepted definition is available.
 - The operating system exists because they are a reasonable way to solve the problem of creating a usable computing system.

Components of an operating system

- What's the part of the operating system?
 - Various people, company or organization has different opinions.
 - Microsoft insisted that web browser and media player were parts of the operating system.
- In this course, we will focus on the **kernel** of the operating system.
 - The kernel provides the lowest level of abstraction layer for the resources (especially memory, processors and I/O devices). It includes (but is not limited to) the following components
 - **CPU manager**
 - **Memory Manager**
 - File System
 - Device Manager

History of Operating System

- To see what operating systems are and what they do, we will consider how they have developed over past 45 years.
- Operating system runs on specific hardwares.
 - We cannot understand operating system without some knowledge of underlying hardware.
- So, we will trace the evolution of computer systems and their operating systems to identify the common elements of operating systems.
 - Mainframes and mini-computers
 - Mainframe - IBM System z9
 - Mini-computer - IBM System i
 - Desktop computers
 - Apple II, Macintosh
 - IBM Personal Computer
 - Embedded computers

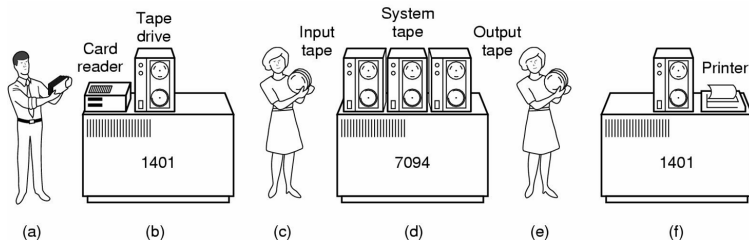
Operating systems for mainframes and minicomputers

- Mainframes and minicomputers usually have dedicated operating systems.
 - **zOS** is the operating system for IBM System z9.
 - **OS/400** is the operating system for IBM System i.
- They evolved from simple **batch system**, to **multiprogramming system** and to **time-sharing system**.

Batch systems(1/4)

- The computer runs one and only one application at a time.
- Batching similar jobs
 - Automatically transfers control from one job to another.
- First rudimentary operating system.

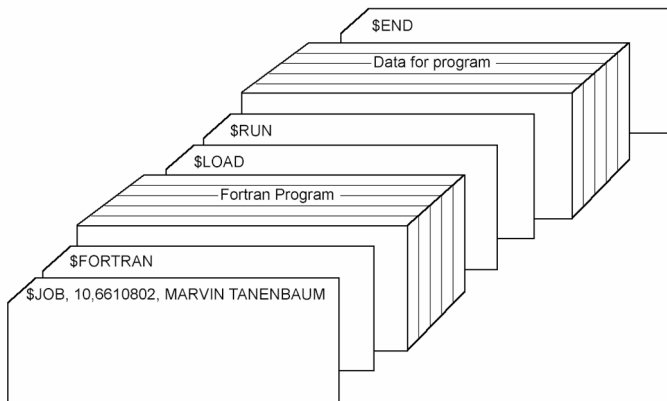
Batch systems(2/4)



- (a) Programmers bring cards to IBM 1401 (b) 1401 reads batch of jobs into tape.
- (c) Operator carries input tape to IBM 7094 (d) 7094 does computing.
- (e) Operator carries output tape to IBM 1401. (f) 1401 prints output.

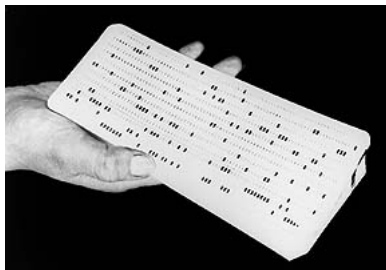
Batch systems(3/4)

- A typical job:



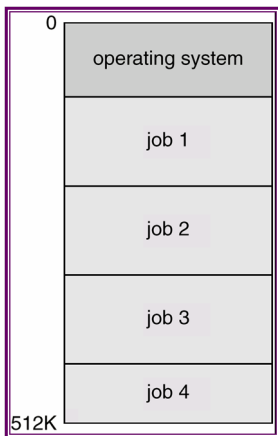
Batch system(4/4)

- A punch card

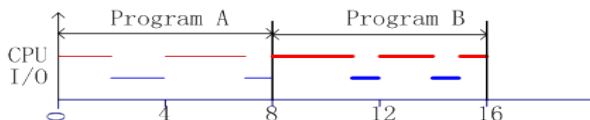


Multiprogramming systems

- Several jobs are kept in main memory at the same time, and the CPU is multiplexed among them.



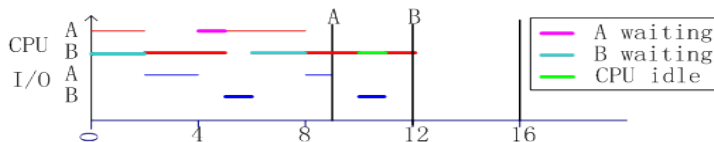
Batch v.s. Multiprogramming system



Uniprogramming

Program A and B complete at time 16

CPU idle: .357 I/O idle: .643



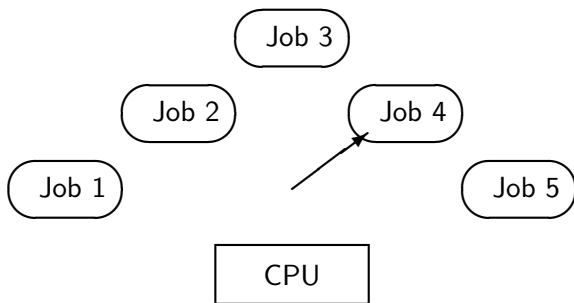
Multiprogramming

Program A and B complete at time 12

CPU idle: .083 I/O idle: .583

Time-sharing systems

- The CPU is multiplexed among several jobs that are kept in memory and on disk (the CPU is allocated to a job only if the job is in memory).
 - Designed for interactive computing which requires quick response time.



Operating systems for desktop computers

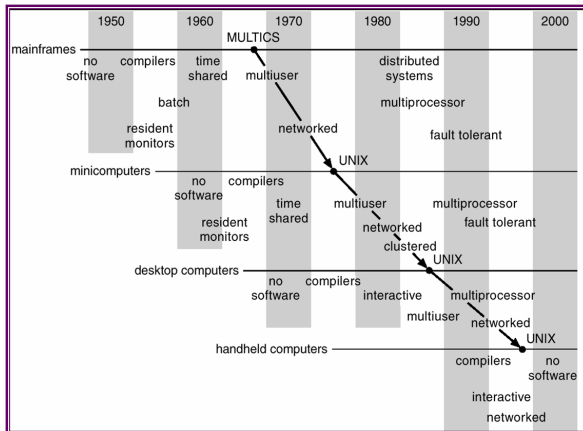
- Operating systems for these computers have benefited in several ways from the development of the operating systems for mainframes.
 - Apple Macintosh
 - Microsoft MS-DOS, Windows 9x/NT
 - IBM OS/2
 - Unix
 - **BSD** (Berkeley Software Distribution) Unix, including **FreeBSD**, **OpenBSD** and **NetBSD**
 - Solaris by Sun microsystem (acquired by Oracle on 2009/04/20)
 - Apple macOS
 - Linux
 -

Operating systems for embedded systems

- Embedded computer is often used as a control device in a dedicated application such as industrial control systems. Usually, they have limited resources:
 - Slow processor, limited memory.
 - Small or even no display screen.
 - Limited power supply, etc
- Some control devices have time requirement, i.e., **Real time**
 - **Hard** real time, actions absolutely **MUST** occur at a certain moment.
 - **Soft** real time, missing an occasional deadline is acceptable.
- Examples
 - Windriver vxWorks
 - Linux
 -

Features migration

- History repeats itself.



Questions

- Any questions?

